

NICOLETTE MASHABA

Software Engineer — Full-Stack Developer — Microsoft Certified Azure Developer

Email: nene171408@gmail.com | Phone: +27 63 152 6795

LinkedIn: [linkedin.com/in/nicolette-mashaba](https://www.linkedin.com/in/nicolette-mashaba) | GitHub: github.com/NickiMash17

Portfolio: nicmash-porfolio.vercel.app | Location: Polokwane, South Africa

PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in full-stack web development, mobile application development, cloud computing, and artificial intelligence. Microsoft Certified Azure Developer with proven success building production-ready applications using React, Node.js, .NET Core, Flutter, and Python. Delivered measurable results including 25 percent AI model accuracy improvement and cross-platform mobile apps with 80 percent test coverage. Strong problem-solver with Agile development experience, technical training skills, and passion for creating scalable solutions. Seeking entry-level software engineering role to contribute technical expertise and grow with an innovative team.

TECHNICAL SKILLS

Programming Languages	JavaScript, TypeScript, Python, C Sharp, Dart, SQL, HTML5, CSS3
Frontend Technologies	React, Next.js, Tailwind CSS, Bootstrap, Vite, Recharts, Responsive Web Design
Backend Technologies	Node.js, Express.js, ASP.NET Core, .NET Framework, FastAPI, RESTful API, GraphQL
Mobile Development	Flutter, Dart, React Native, BLoC Pattern, Provider, Cross-Platform Development
Cloud Platforms	Microsoft Azure, Azure App Service, Azure Functions, Firebase, Google Cloud Platform
Database Systems	MongoDB, Microsoft SQL Server, PostgreSQL, MySQL, Firestore, Redis
DevOps and Tools	Git, GitHub, Docker, CI/CD Pipelines, Terraform, Azure DevOps, Postman
AI and Machine Learning	OpenAI API, Natural Language Processing, Prompt Engineering, Model Training, TensorFlow
Software Development	Agile Scrum, Test-Driven Development, Unit Testing, Integration Testing, Code Review
Architecture Patterns	Microservices, Clean Architecture, MVC, MVVM, Repository Pattern, Dependency Injection

PROFESSIONAL EXPERIENCE

Technical Trainer — Techbridle Foundation (Volunteer)	January 2026 – Present
– Deliver software development training to aspiring developers, teaching HTML, CSS, JavaScript, React, and Node.js – Mentor students in programming fundamentals, debugging techniques, and software engineering best practices – Create hands-on coding exercises and real-world projects to build practical development skills – Provide code reviews and technical feedback to improve student code quality and problem-solving abilities – Guide students in Git workflow, version control, and collaborative development practices	
Software Engineering Intern — Coast IT Proprietary Limited	November 2025 – December 2025
– Developed mobile application features using Flutter for Android and iOS, delivering cross-platform functionality – Collaborated with development team in Agile Scrum environment, participating in daily standups, sprint planning, and code reviews – Implemented bug fixes and feature enhancements, improving application stability and user experience – Contributed to quality assurance processes, writing unit tests and conducting manual testing – Worked with Git version control for code management and team collaboration	

- AI Model Trainer — Outlier AI

October 2024 – December 2024

 - Trained and refined artificial intelligence models, improving accuracy by 25 percent through prompt optimization
 - Created structured datasets for natural language processing model training and validation
 - Developed annotation workflows that reduced engineering team workload by 10 hours per week
 - Collaborated with machine learning engineers to identify model weaknesses and improvement opportunities
 - Documented testing procedures and evaluation metrics for AI model performance tracking

PROJECTS

- EmpowerAI — Digital Economic Twin Platform2024
 - GitHub: github.com/NickiMash17/EmpowerAI
 - Led development of AI-powered career simulation platform using React, Node.js, Python, and OpenAI API
 - Architected full-stack application with microservices design pattern and RESTful API architecture
 - Implemented natural language processing for CV analysis and interview preparation features
 - Built real-time data visualization dashboards using React, Recharts, and Tailwind CSS
 - Deployed application on Microsoft Azure with MongoDB Atlas database and automated CI/CD pipeline
 - Technologies: TypeScript, React, Node.js, Express, Python, FastAPI, OpenAI API, MongoDB, Azure, Docker

- FitQuest — Mobile Wellness Application2024
 - GitHub: github.com/NickiMash17/fitquest-app
 - Built cross-platform mobile application for Android, iOS, and Web using Flutter and Dart
 - Implemented Firebase backend with Authentication, Firestore database, and Cloud Functions
 - Designed Clean Architecture with BLoC state management and repository pattern for scalability
 - Created custom animations using Rive for enhanced user experience and engagement
 - Achieved 80 percent test coverage with comprehensive unit tests and widget tests
 - Technologies: Flutter, Dart, Firebase, BLoC, Rive, Git

- AI Compliance Platform — AIMS Hackathon Winner2025
 - GitHub: github.com/NickiMash17/AIMS-Firefly
 - Won Best Team in Application and Visualisation at AIMS Hackathon 2025
 - Developed AI-powered compliance detection system using React, .NET Core, and OpenAI API
 - Built real-time dashboard with live notifications and interactive data visualizations
 - Implemented natural language processing pipelines for document analysis and risk detection
 - Deployed full-stack solution on Microsoft Azure with SQL Server database
 - Technologies: React, .NET Core, C Sharp, Azure, SQL Server, OpenAI API

EDUCATION

- 2026 (Expected)Occupational Certificate in Software Engineering, NQF Level 6CTU Training Solutions, Polokwane, South AfricaRelevant Coursework: Data Structures, Algorithms, Software Design, Database Management
- 2024IT Programming Foundation, NQF Level 4CTU Training Solutions, Polokwane, South AfricaAchievement: Top Performer 2023

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate (AZ-204) — Issued 2024

- **Microsoft Certified: Azure Data Fundamentals (DP-900)** — Issued 2024
- **J.P. Morgan Software Engineering Job Simulation** — Completed 2025
- **Techbridle Foundation Software Development Bootcamp** — Completed 2024

AWARDS AND ACHIEVEMENTS

- Winner, AIMS Hackathon 2025 — Best Team in Application and Visualisation
- Top Performer, CTU Programming Foundation — 2023
- Techbridle Foundation Bootcamp Graduate with Distinction

ADDITIONAL EXPERIENCE

Junior Navigation Officer — South African Navy January 2015 – July 2021

- Optimized operational processes, achieving 10 percent fuel cost reduction through efficient voyage planning
- Conducted technical equipment audits, reducing operational downtime by 15 percent
- Trained and mentored junior officers in navigation procedures and safety protocols
- Demonstrated strong analytical thinking, attention to detail, and ability to work under pressure